

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2024-2025
FULL MARKS: 75

CSE 4513: Software Engineering and Object-Oriented Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Although Line of Code (LOC) is the simplest metric for software size estimation, it is not widely used in practice. Do you agree? Justify your answer. 5
(CO1)
(PO1)
b) As a software engineer in a software company, you are assigned to develop an online payment system called VecnaPay. The team writes short requirements on cards. They discuss the details with stakeholders in meetings. Acceptance tests are used to check whether the features work as expected. Before finalizing the requirements document, you review the requirements to make sure they match stakeholder needs, requirements should not conflict with each other, include all necessary system functions such as payment validation and refund handling, are feasible to implement, and can be tested. 6 + 4
(CO1)
(PO1)
 - i. Explain how the 3C approach can be used to write user stories for the VecnaPay system.
 - ii. From the scenario, identify and explain any two checks that should be performed on the requirements to ensure good quality.
- c) Consider a software project with the following parameters: EI = 45, EO = 30, EQ = 25, ILF = 08, and EIF = 05. Assume all weighting factors are average. In addition, the system requires moderate performance, significant end-user efficiency, average distributed data processing, moderate data communication, and significant transaction rates. All other General System Characteristics (GSCs) are essential. Compute the Function Points of the system using Function Point Analysis. 8
(CO2)
(PO2)
2. The Hawkins Lab, located in the city of Hawkins, is developing a software system to monitor and analyze the city because stranger things are happening in different areas. Due to unclear and frequently changing requirements, the team follows Agile Scrum. In the early phase of the project, a prototype is created only to understand user requirements, and it will not be part of the final system. The team works in short sprints, tracks progress using Scrum artifacts, and places strong emphasis on both functional and non-functional quality before releasing features.
 - a) Identify the most suitable prototyping model for the given scenario and justify your answer with appropriate reasoning. 5
(CO1)
(PO1)
 - b) At the end of each sprint, the Hawkins Lab Scrum team conducts a Sprint Retrospective. Describe why the Sprint Retrospective is conducted and what the team aims to achieve through it. 5
(CO1)
(PO1)
 - c) The Hawkins Lab system should meet strict non-functional requirements such as reliability, security, and performance. Explain the importance of the "Definition of Done", particularly in ensuring that non-functional requirements are satisfied. 5
(CO1)
(PO1)
 - d) During a sprint, the Scrum team at Hawkins Lab notices that the actual burndown chart temporarily increases on some days instead of always decreasing. Explain the probable reasons for this situation and discuss whether it necessarily indicates a problem in the Scrum process. 6
(CO1)
(PO1)

- e) While planning the sprint, a team member suggests converting story points into hours for easier estimation. Explain whether this practice is appropriate in Agile Scrum. 5
(CO1)
(PO1)
3. a) "General requirements such as friendliness should be avoided." Discuss this statement in the context of software requirements. 5
(CO1)
(PO1)
- b) A newly deployed university portal is difficult for first-time users to understand, responds slowly during common tasks, and crashes when users make small input mistakes. From the user interface evaluation attributes, identify three relevant attributes that are not being satisfied and justify your choices. 6
(CO2)
(PO2)
- c) Write short notes on the following topics. Each note should include a brief explanation and one relevant example. 5 × 3
(CO1)
(PO1)
- i. YAGNI principle
 - ✓ ii. Mean Time to Failure (MTTF)
 - iii. KISS principle